A/C COMPRESSOR CLUTCH

DESCRIPTION

The A/C compressor clutch assembly consists of a stationary electromagnetic A/C clutch field coil, a pulley bearing and pulley assembly, and a clutch plate (Fig. 1). These components provide the means to engage and disengage the A/C compressor from the engine accessory drive belt.

The A/C clutch field coil and the pulley bearing and pulley assembly are both retained on the nose of the A/C compressor with snap rings. The clutch plate is splined to the compressor shaft and secured with a bolt.

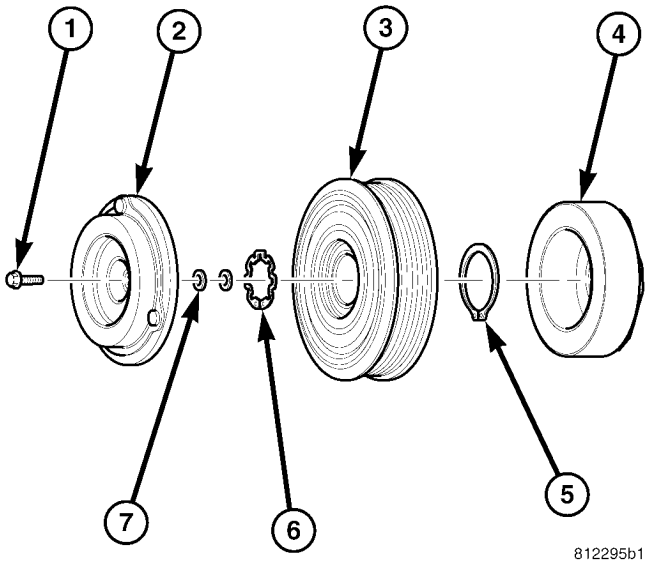

Fig. 1 A/C Compressor Clutch

1 - BOLT
2 - CLUTCH PLATE
3 - PULLEY AND BEARING
4 - FIELD COIL
5 - SNAP RING
6 - SNAP RING
7 - SHIMS

OPERATION

The A/C compressor clutch components provide the means to engage and disengage the A/C compressor from the engine accessory drive belt. When the electromagnetic A/C clutch field coil is energized, it magnetically draws the clutch plate into contact with the clutch pulley and drives the compressor shaft. When the coil is not energized, the pulley freewheels on the clutch hub bearing, which is part of the pulley.

A/C clutch engagement is controlled by the following:

- A/C-heater mode control
- A/C clutch relay
- A/C high pressure switch
- A/C low pressure switch
- Powertrain control module (PCM)

The A/C compressor clutch components cannot be repaired and, if faulty or damaged, they must be replaced.

DIAGNOSIS AND TESTING

A/C COMPRESSOR CLUTCH COIL

The A/C compressor clutch coil electrical circuit is controlled by the powertrain control module (PCM) through the A/C clutch relay. Begin testing of a suspected compressor clutch coil problem by performing the preliminary checks.

PRELIMINARY CHECKS

(1) If no diagnostic trouble codes (DTCs) are found in the powertrain control module (PCM), go to Step 2. If any DTCs are found, repair as required.

(2) If the A/C compressor clutch still will not engage, verify the refrigerant charge level (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - DIAGNOSIS AND TESTING - REFRIGERANT SYSTEM LEAKS). If the refrigerant charge level is OK, go to Step 3. If the refrigerant charge level is not OK, adjust the refrigerant charge as required.

(3) On models equipped with the 2.4L engine, if the A/C compressor clutch still will not engage, disconnect the wire harness connector for the A/C pressure transducer and check for battery current at the connector with the engine running and the A/C-heater control set to the A/C mode. If OK, go to TESTS. If not OK, refer to the Body Diagnostic Procedures to perform further diagnosis. On models equipped with the 4.0L engine, go to TESTS.

TESTS

(1) Verify the battery state of charge (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/BATTERY - DIAGNOSIS AND TESTING).

(2) Connect an ammeter (0 to 10 ampere scale selected) in series with the clutch coil feed terminal. Connect a voltmeter (0 to 20 volt scale selected) to measure voltage across the battery and the clutch coil.

(3) With the A/C-heater control in the A/C mode and the blower motor at low speed, start the engine and allow it to run at a normal idle speed.

(4) The A/C compressor clutch should engage immediately, and the clutch coil supply voltage should be within two volts of the battery voltage. If the coil supply voltage is OK, go to Step 5. If the coil supply voltage is not within two volts of battery voltage, test the clutch coil feed circuit for excessive voltage drop and repair as necessary. If there is no voltage reading at the clutch coil, use a DRB III ® scan tool and refer to appropriate diagnostic informa-